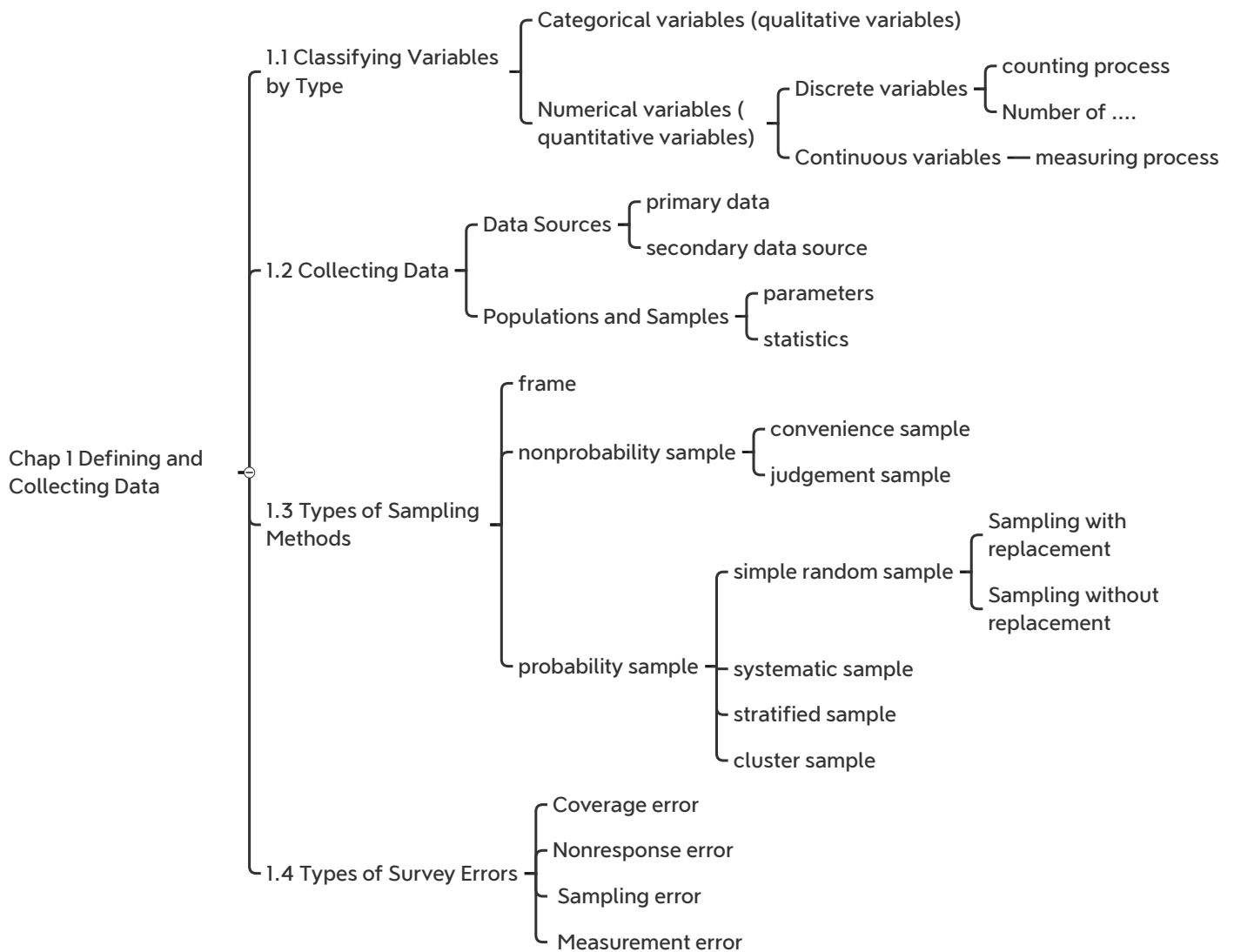
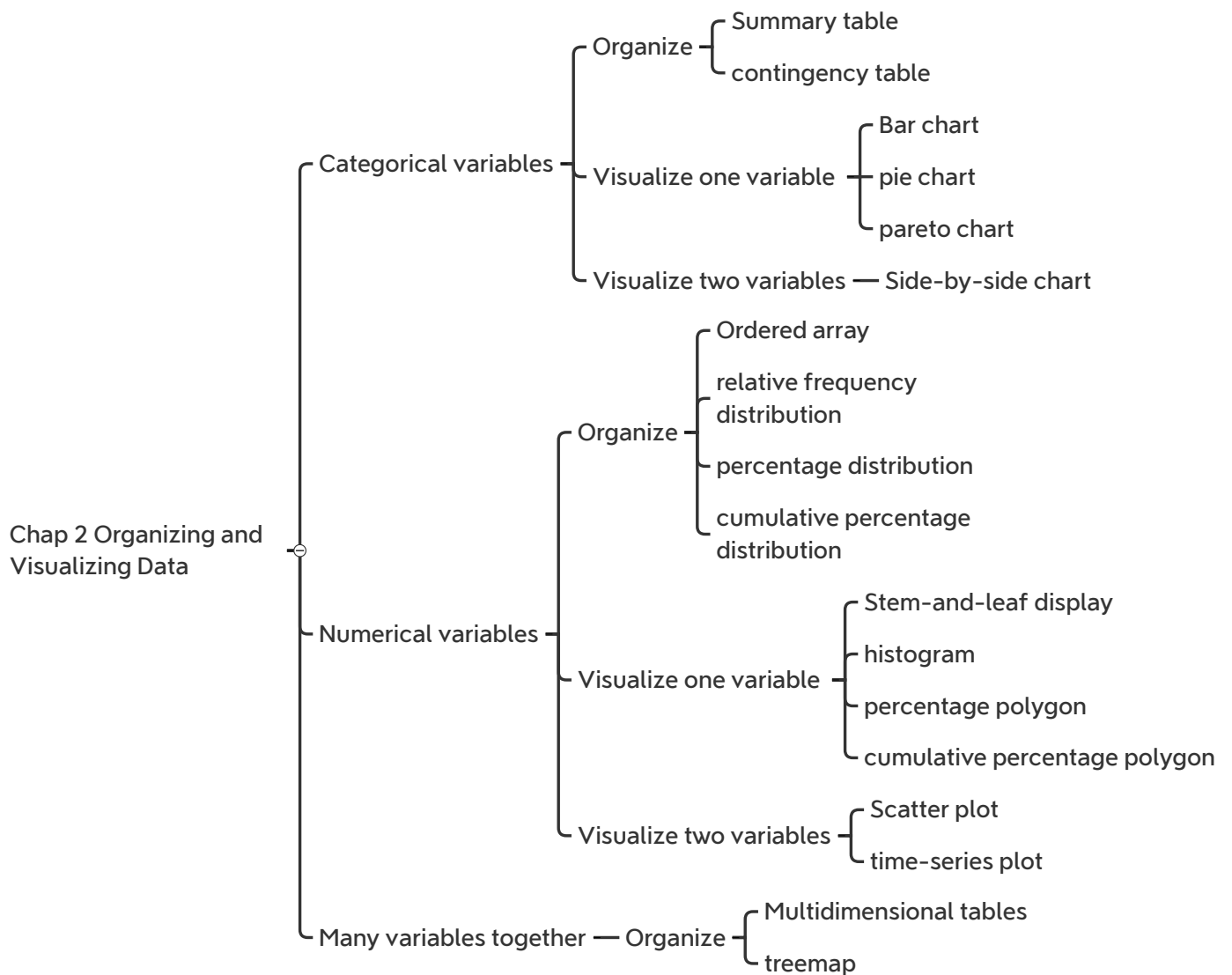
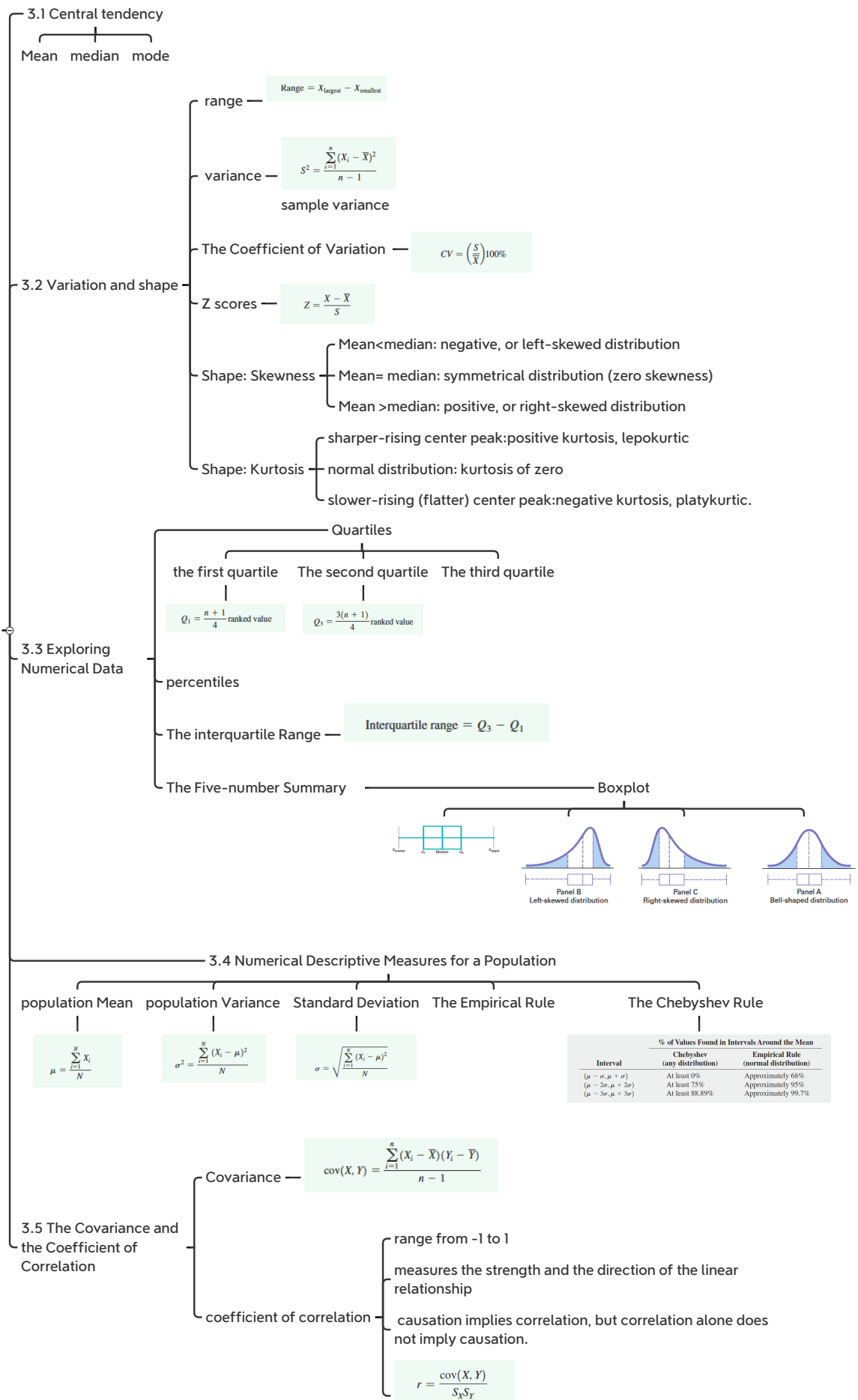


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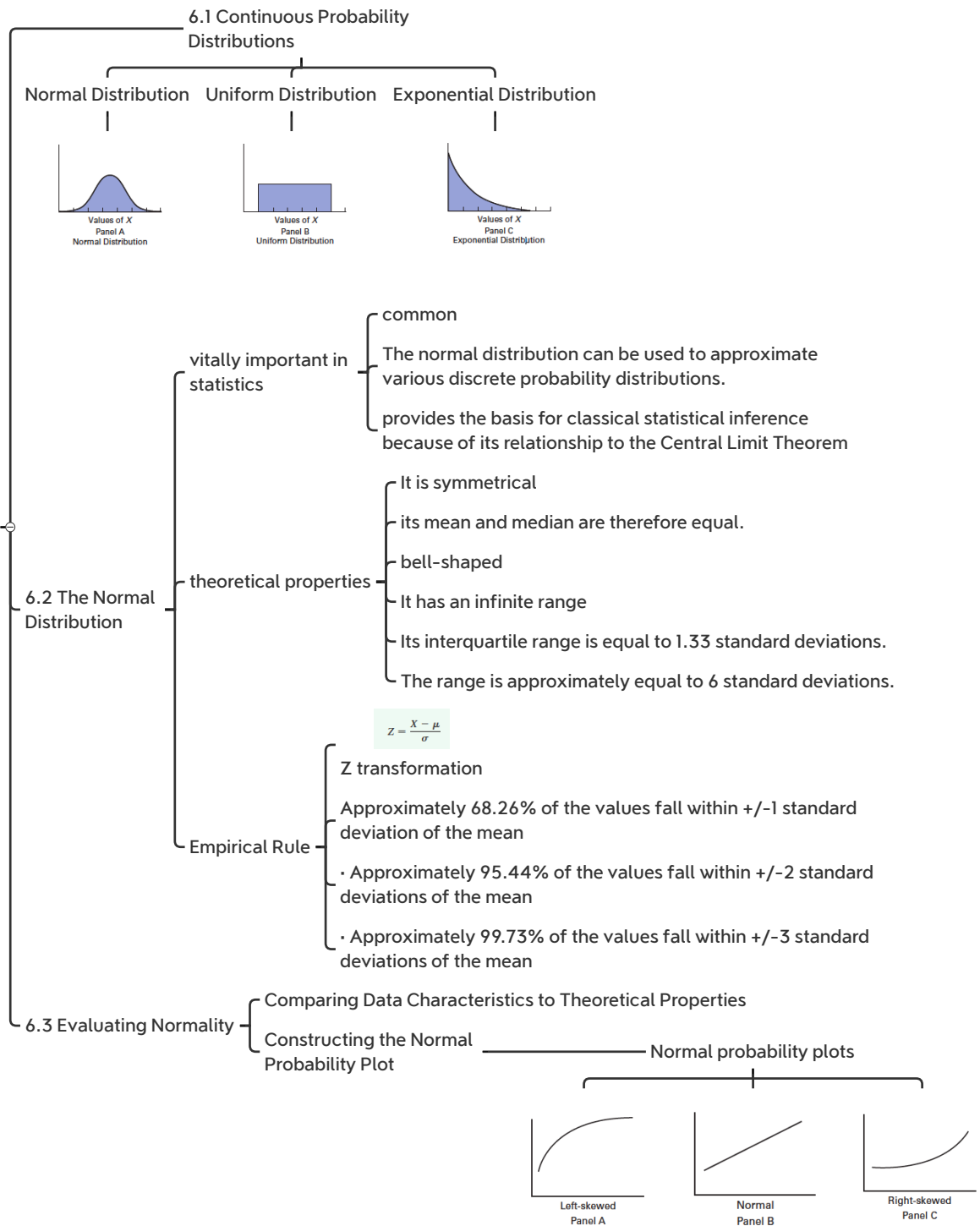


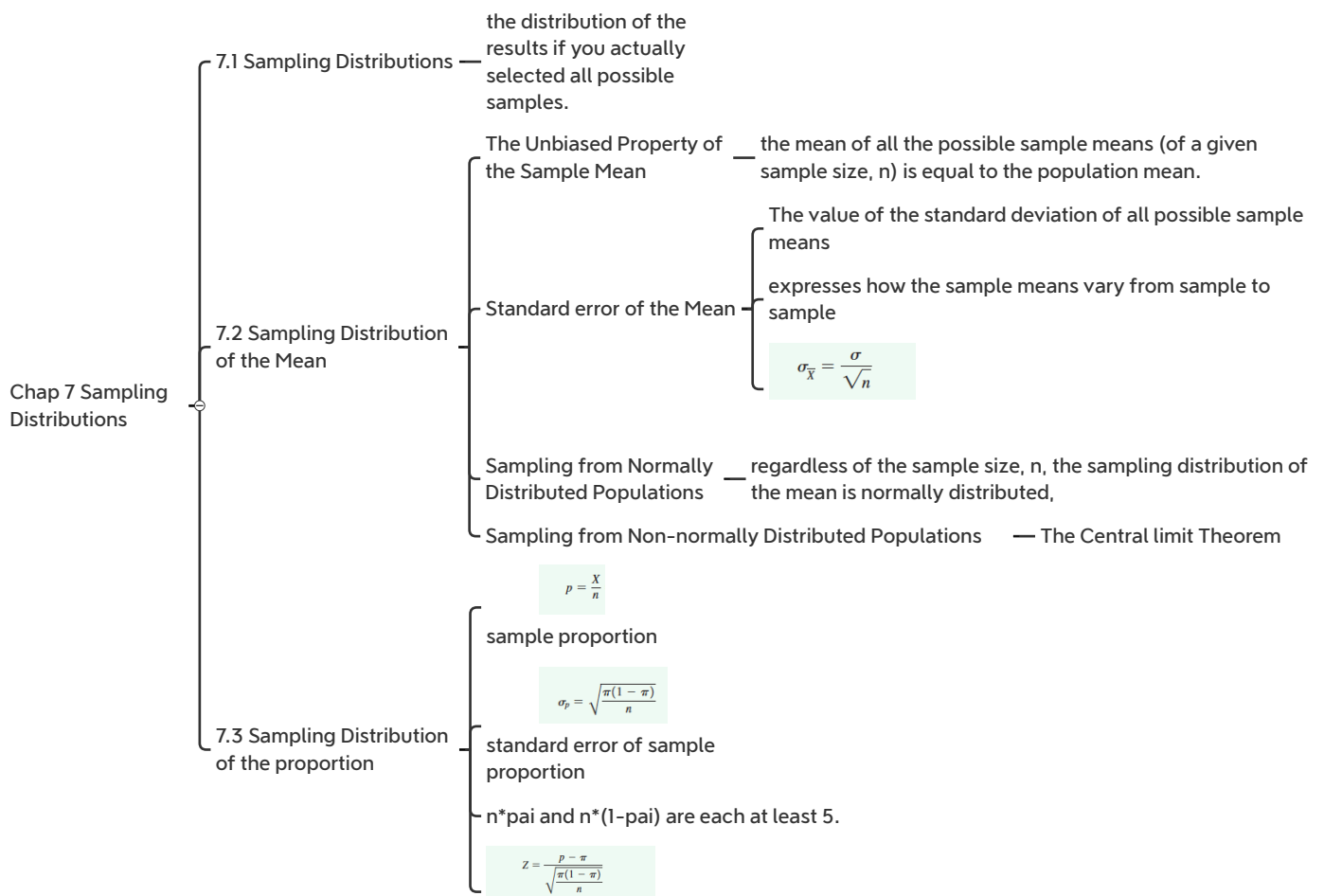


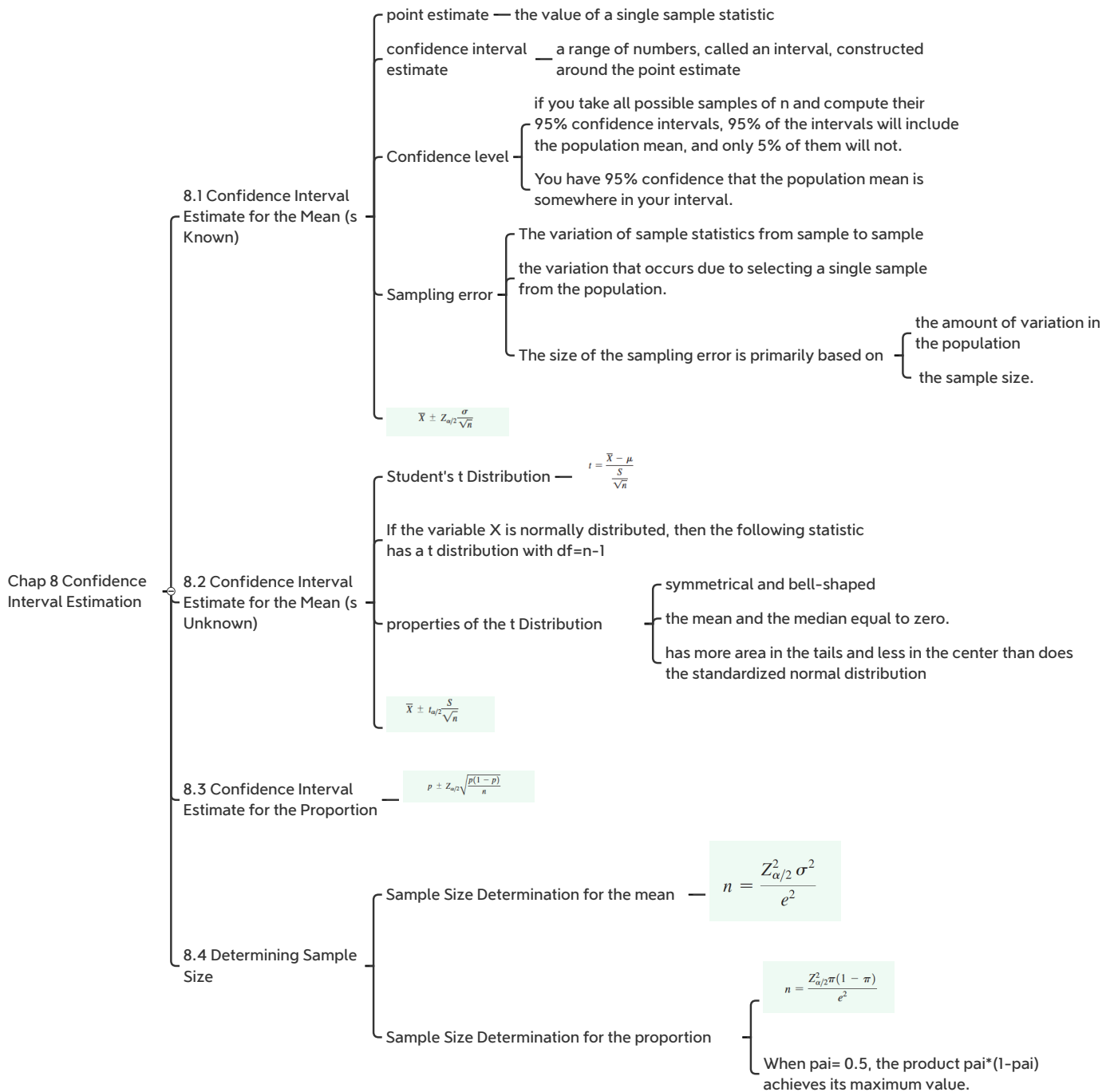
Chapter 3 Descriptive Statistics Methods



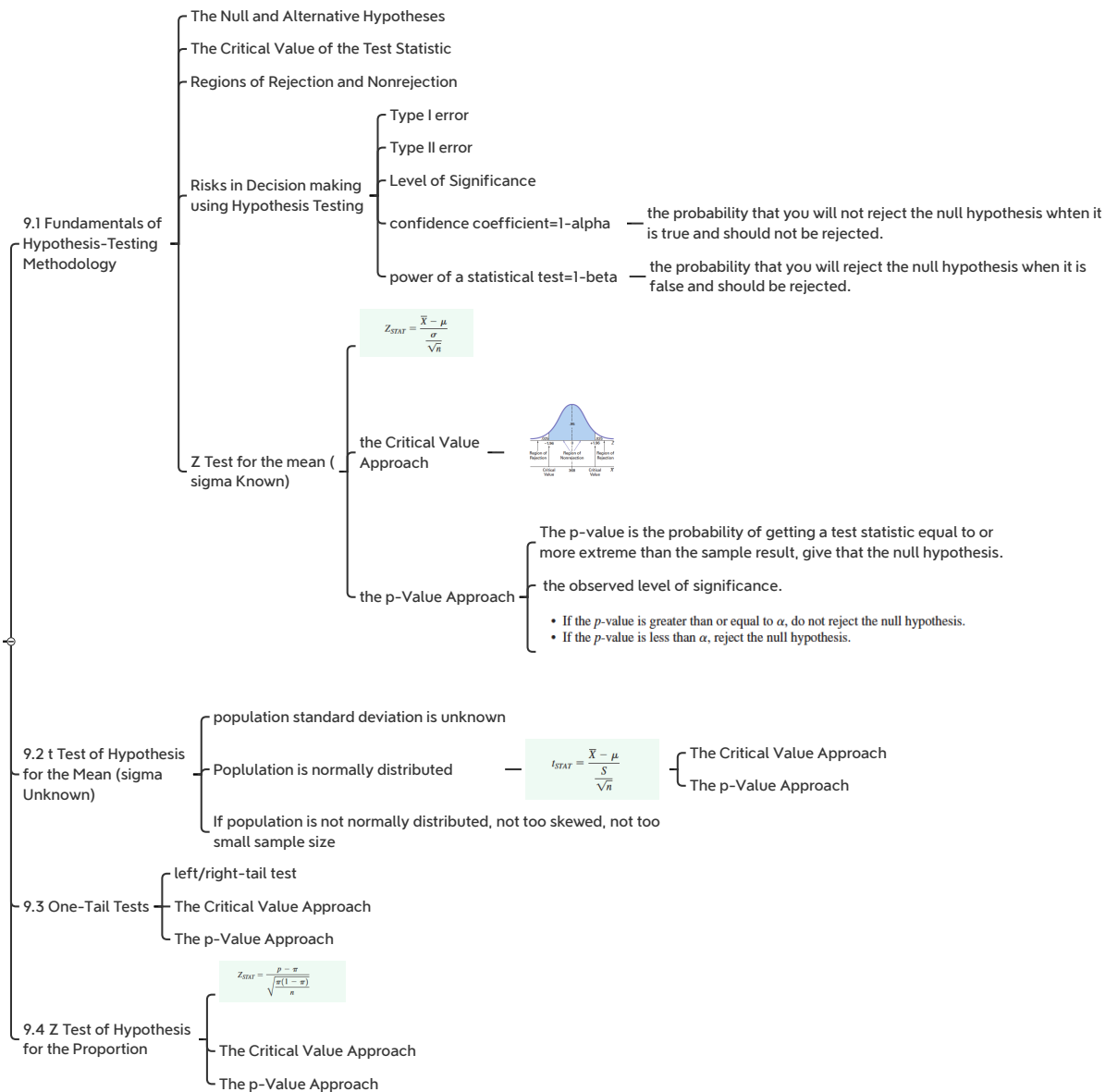
Chap 6 The Normal Distribution







Chap 9 Fundamentals of Hypothesis Testing: One-Sample Tests



- 10.1 Comparing the Means of Two Independent Populations — Pooled-Variance t Test — for the Difference Between Two Means — EXCEL

assume that the populations are normally distributed, with equal variances

T test

$$t_{STAT} = \frac{\bar{D} - \mu_D}{\frac{S_D}{\sqrt{n}}}$$

EXCEL

- 10.2 Comparing the Means of Two Related Populations — Paired t Test —

- 10.3 Comparing the Proportions of Two Independent Populations — Z Test for the Difference Between Two Proportions

$$Z_{STAT} = \frac{(p_1 - p_2) - (p_1 - p_2)}{\sqrt{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2} \quad p_1 = \frac{X_1}{n_1} \quad p_2 = \frac{X_2}{n_2}$$

- 10.4 F Test for the Ratio of Two Variances — assume that each population is normally distributed

$$F_{STAT} = \frac{S_1^2}{S_2^2}$$

EXCEL

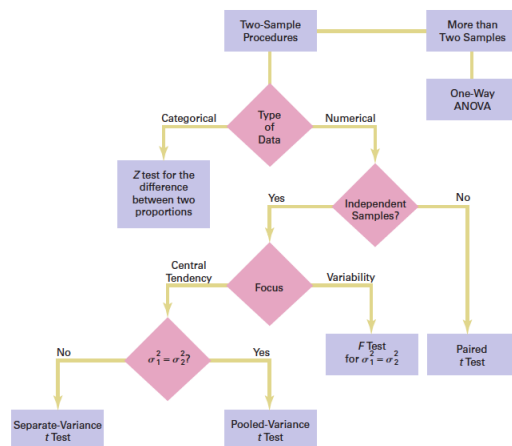
$$H_0: \mu_1 = \mu_2 = \dots = \mu_c$$

H_1 : Not all μ_j are equal (where $j = 1, 2, \dots, c$).

- 10.5 One-Way ANOVA

- F Test for Differences among More Than Two Means — EXCEL
- Levene Test for homogeneity of Variance — EXCEL SPSS
- Multiple Comparisons: LSD — SPSS

Source	Degrees of Freedom	Sum of Squares	Mean Square (Variance)	F
Among groups	$c - 1$	SSA	$MSA = \frac{SSA}{c - 1}$	$F_{STAT} = \frac{MSA}{MSW}$
Within groups	$n - c$	SSW	$MSW = \frac{SSW}{n - c}$	
Total	$n - 1$	SST		



Chapter 11 Chi-Square Tests

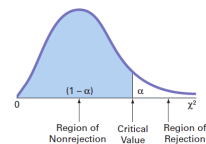
11.3 Chi-Square Test of Independence

test of independence for two categorical variables.

H_0 : The two categorical variables are independent

H_1 : The two categorical variables are dependent

$$\chi^2_{STAT} = \sum_{all\ cells} \frac{(f_o - f_e)^2}{f_e}$$



Reject H_0 if $\chi^2_{STAT} > \chi^2_{\alpha}$;
otherwise, do not reject H_0 .

$$f_e = \frac{\text{Row total} \times \text{Column total}}{n}$$